Extended iStar Language

1 Model Structure

An extended iStar model is

$$\mathcal{G} = \langle Actor, IE, \mathcal{R}, \kappa_G, \tau \rangle,$$

where

$$IE = Goal \uplus Task \uplus Quality, \qquad \mathcal{R} = \mathbf{??}$$

Con$\uplus$*Dep*. Each intentional element has a unique owner

$$owner : IE \to Actor.$$

Refinement links are

$$\mathbf{??}$$

(Goal$\cup$*Task*) $\times$ (*Goal* $\cup$ *Task*), where $(c, p) \in \mathbf{??} meansthatcrefinesparent$p.$Define Ch(p) =$ $\{c \mid (c, p) \in \mathbf{??}$5267309

, Root={x$\in$ $Goal \cup Task \mid \nexists p : (x, p) \in \mathbf{??}$5267309

.

For every non-leaf parent, all incoming refinement links have one common type:

$$rtype : \{p \mid Ch(p) \neq \varnothing\} \to \{AND, OR\}.$$

Contribution and dependency links are retained as

$$Con \subseteq (Goal \cup Task) \times Quality, \qquad Dep \subseteq IE \times IE,$$

with contribution labels

$$ctype : Con \to \{Make, Help, Hurt, Break\}.$$

The paper does not give a complete quantitative propagation rule for contribution labels; therefore they are structural in this core semantics unless an explicit contribution policy is supplied.

2 State-Dependent Specifications

Let $\mathcal{A}$ be the shared ACL model and Σ_A its admissible states. State-dependent specifications are attached only to leaves:

$$cond : \{g \in Goal \mid Ch(g) = \varnothing\} \to BoolExpr(\mathcal{A}),$$

$$pre_G, post_G : \{t \in Task \mid Ch(t) = \varnothing\} \to BoolExpr(\mathcal{A}).$$

Missing specifications denote *true*. Goal temporal types are

$$\tau : Goal \to \{Achieve, Maintain, Sustain\}.$$

3 Marking Domain

Following the paper, intentional values use

$$\Delta = \{\bot, ?, \top\},$$

where $\bot$ is violated, ? is undetermined, and $\top$ is satisfied. A task additionally has a pending control value P:

$$\Delta_T = \{?, P, \top\}.$$

Thus the task progression is

$$? \xrightarrow{pre_G(t)} P \xrightarrow{post_G(t)} \top.$$

4 Leaf Evaluation

For $\sigma_i \in \Sigma_A$, a leaf goal has structural value

$$val_i^S(g) = \begin{cases} \top, & \sigma_i \models cond(g), \\ \bot, & \text{otherwise.} \end{cases}$$

For a leaf task, with $task_{-1}(t) =?$,

$$task_i(t) = \begin{cases} \top, & task_{i-1}(t) = \top, \\ \top, & task_{i-1}(t) = P \wedge \sigma_i \models post_G(t), \\ P, & task_{i-1}(t) = P, \\ P, & task_{i-1}(t) =? \wedge \sigma_i \models pre_G(t), \\ ?, & \text{otherwise.} \end{cases}$$

Its structural contribution is

$$val_i^S(t) = \begin{cases} \top, & task_i(t) = \top, \\ ?, & task_i(t) \in \{?, P\}. \end{cases}$$

Hence a task never directly produces $\bot$, as stated in the paper.

5 Structural Propagation

For a non-leaf x with children already evaluated,

$$val_i^S(x) = \begin{cases} \top, & rtype(x) = AND \wedge \forall y \in Ch(x) : val_i(y) = \top, \\ \bot, & rtype(x) = AND \wedge \exists y \in Ch(x) : val_i(y) = \bot, \\ \top, & rtype(x) = OR \wedge \exists y \in Ch(x) : val_i(y) = \top, \\ \bot, & rtype(x) = OR \wedge \forall y \in Ch(x) : val_i(y) = \bot, \\ ?, & \text{otherwise.} \end{cases}$$

For a dependency $(x, d) \in Dep$, the dependee value is propagated to the depender-side dependency occurrence:

$$val_i^S(x \mid d) = val_i(d).$$

6 Temporal Goal Semantics

Initialize $val_{-1}(g) = ?$ and define

$$val_i(g) = Temp_{\tau(g)}\big(val_{i-1}(g), val_i^S(g)\big).$$

The three temporal operators are

$$Temp_{Achieve}(p, s) = \begin{cases} \top, & p = \top \vee s = \top, \\ ?, & \text{otherwise}, \end{cases}$$

$$Temp_{Maintain}(p, s) = \begin{cases} \bot, & p = \bot \vee s = \bot, \\ \top, & s = \top, \\ ?, & \text{otherwise}, \end{cases}$$

$$Temp_{Sustain}(p, s) = \begin{cases} \bot, & p = \bot, \\ \bot, & p = \top \wedge s = \bot, \\ \top, & s = \top, \\ ?, & \text{otherwise}. \end{cases}$$

This makes the paper's persistence statements explicit: Achieve preserves satisfaction, Maintain preserves detected violation, and Sustain preserves violation after a previously achieved goal is broken.

7 Intentional Marking and Scenario Verdict

At step i, the intentional marking is

$$\mu_i : IE \to \Delta,$$

using val_i for goals and the structural task value for tasks. For a complete system scenario with final marking μ_n:

$$Conformant(\pi) \iff \forall g \in Root \cap Goal : \ \mu_n(g) = \top,$$

$$Risky(\pi) \iff \exists g \in Root \cap Goal : \ \mu_n(g) = \bot.$$